



Summer water temperature

A report prepared on behalf of Wye Salmon Association (WSA)

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A short discussion paper on key drivers of summer water temperature

Introduction

The summer of 2026 saw extended periods of dangerously elevated water temperatures on the main stem of the Wye. This note attempts to explore the factors which influence this.

Heat fluxes

The temperature of a river is driven by heat flows. These can be (a) conductive: to or from the air and riverbed, (b) radiative: solar input and long-wavelength radiative losses, and (c) mass transfer: ingress of groundwater and surface evaporation. Depending on the location and time, different mechanisms may dominate. Simple conduction to or from the air does occur but is generally relatively weak. Summer heating is dominated by the sun.

Solar radiation

At the summer solstice (21-June), the mid-day sun reaches about 60° elevation in Herefordshire. At ground level, the incident solar radiation is 850-900 W/m². The range relates to air clarity, with the higher value being applicable to crisp, clear conditions and the lower value in hazy conditions. If there is cloud cover, the numbers drop significantly; but 2026 saw relatively little cloud cover most days. At 60° elevation angle, 90-95% of this radiation gets into the water column where it is absorbed directly, or indirectly via heating of the gravel. Earlier and later in the day, solar radiation reduces according to a cosine relationship. At low elevations atmospheric scattering further reduces incident radiation. Also, there is less penetration into the water column at lower angles. Earlier and later in the year, heating rates are further reduced by the lower maximum elevation angle of the sun.

Key point 1: solar radiation is the dominant day-time heat flux in summer.

If 800W (90% of 850W) is absorbed in a water column 1200mm deep (Wye at Ross), the water warms at 0.57°C/hour.

Key point 2: the smaller the depth of water is in the river, the faster the sun warms it up.

Shading

Solar radiation is subject to shading (a) by riparian vegetation, and (b) by the channel itself. Both effects can be significant in smaller streams but are largely insignificant on the main stem of the Wye. It is simple geometry that a 10m high tree does not provide significant shade to a 50m wide river. Far less does 3m of bank above the water surface. But in a 4-5m wide stream (Garren at Packsaddle Bridge), the same trees and bank height provide a very significant amount of shading. Moreover, the degree of such shading depends on the flow direction of the stream. An east-west flowing stream is shaded for a longer period of the day by its channel and riparian vegetation than is a north-south flowing stream.

Google satellite imagery shows that riparian trees completely overshadow large sections of lowland tributaries (e.g. Worm Brook, Garren Brook).

Key point 3: smaller streams can receive significant shading from solar radiation due to channel shape alone. Riparian vegetation can add significantly to this shading. Direction of flow can also be important.

Key point 4: shading is less significant on wide channels (Wye at Ross).

Groundwater infiltration

This is another key difference between small streams and the main stem. Groundwater, feeding from just above the bedrock is at a generally stable temperature, circa 10°C. The effect is to keep streams warmer in the winter and cooler in the summer. But only if the groundwater flow is significant with

respect to the flow in the channel. Thus, for a small tributary stream like the Garren, groundwater flow (which by definition is where the water in Garren comes from) can be very significant, reducing average summer temperatures by several degrees. On the main stem of the Wye however, groundwater input is entirely negligible compared to the huge flow already in the river. In other words, the Wye takes its flow from tributaries rather than from its direct catchment.

Nighttime losses

Temperatures fall back at night, mainly due to a combination of evaporation to air and long-wavelength radiation. These are difficult to calculate because evaporation depends on wind speed, relative humidity, etc. and radiation depends on atmospheric conditions and shading. Conduction to air is typically less significant than evaporation and radiation. Conduction to the bed is usually smaller still.

Sonde data

Main stem in summer

Figure 1 shows a typical 24-hour temperature plot from the EA sonde at Ross Rowing Club in the summer of 2026. Of note during this period were clear skies with minimal cloud cover.

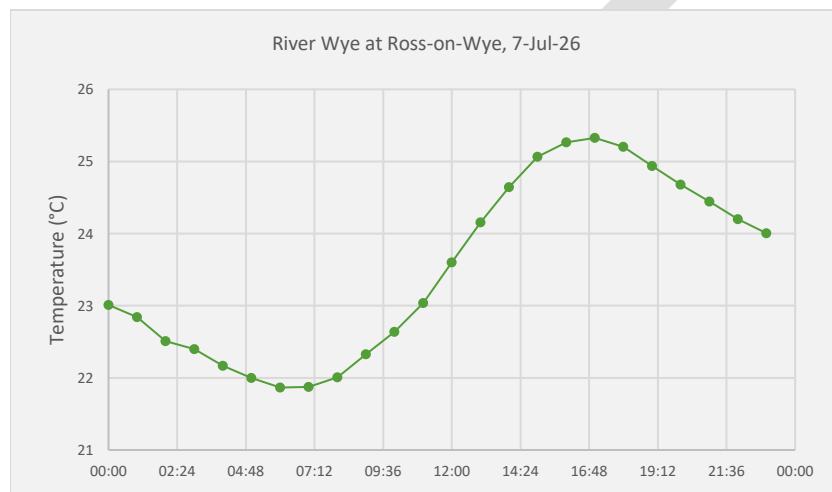


Figure 1: temperature trend plot from the EA sonde at Ross-on-Wye on 07-Jul-26.

NB. times are GMT, so add 1 hour for BST.

The river warmed from about 7:00 GMT (8:00 BST) until about 17:00 GMT (18:00 BST). So, five hours either side of midday, corresponding to elevation angles above about 36° . The maximum rate of increase in temperature, as expected, occurs at 12:00 GMT and equals $0.56^\circ\text{C}/\text{hour}$ (almost exactly as per our theoretical calculation above). The min to max range was 3.46°C .

Nighttime cooling rate was around $-0.25^\circ\text{C}/\text{hour}$, about half of the daytime heating rate. Despite cooling for 14 hours out of 24, compared to heating for just 10 hours, there is no contest here. Indeed, we see that the temperature was on a rising trend, ending the day 1°C hotter than it started out.

Garren in summer

In mid-August 2025, WSA deployed a HydroVu sonde on the Garren Brook at Packsaddle Bridge. Figure 2 shows a similar 24-hour temperature plot for this location. Admittedly, this is a different year (albeit another hot and dry one), and the data is for 6 weeks later in the year, nevertheless the comparison with the Wye at Ross is stark.

Recall that the Garren flows in a general south easterly direction. It is circa 4-5m wide, and the channel is circa 3m below field level. It is also heavily overshadowed by riparian trees. This stream is shallow, circa 300mm, so we might expect more rapid heating than on the main stem – but that isn't what happens.

Heating commenced around 6:00 GMT, not dissimilar to the Wye at Ross. But peak rate of change occurred around 9:00 GMT. That is, when the sun is approximately aligned to the average channel direction, not at midday when the sun is at its zenith. Beyond 9:30 GMT, the rate of change falls progressively. As the sun moves across the sky, the river is increasingly shaded by the channel geometry and riparian vegetation. By 14:30 GMT, shading is essentially total and the water begins to cool.

This brook is only heating for 7 hours each day, compared to 10 hours on the Wye. The peak heating rate at 9:00 GMT is $0.21^{\circ}\text{C}/\text{hour}$, fairly similar to the value on the Wye ($0.23^{\circ}\text{C}/\text{hour}$) at this time of day. However, allowing for the difference in depth, the implication is that the Garren receives, at peak, only about 25% of the solar radiation intensity which hits the Wye. And that is only for a brief period before channel shading reduces it further.

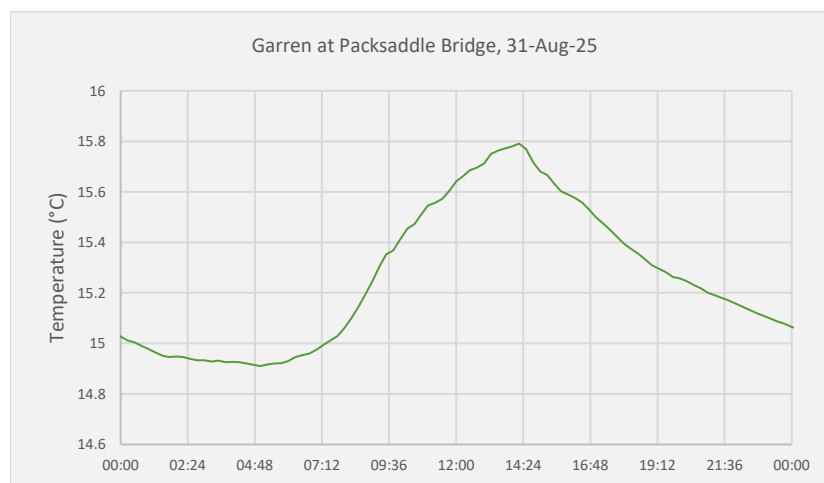


Figure 2: temperature trend plot from the WSA Packsaddle Bridge sonde on 31-Aug-25.

NB. times are GMT, so add 1 hour for BST.

The upshot is that, despite its shallowness, the Garren only warms up by 0.85°C each day and remains, on average, several degrees cooler than the main stem of the Wye in summer. Groundwater input is likely to also be a factor here. According to CS data, maximum recorded temperatures across the Garren and Worm catchments tend to be circa $19\text{--}20^{\circ}\text{C}$; well below the $25\text{--}28^{\circ}\text{C}$ recorded on the main stem.

Conclusion

- Sonde data from the Garren sonde supports our analysis regarding the importance of channel geometry and riparian vegetation for limiting temperate excursions in smaller brooks and streams.
- Average temperatures in small streams are also stabilised by infiltration of cooler groundwater.
- The width of the main stem of the Wye precludes shading, resulting in unmitigated solar radiation fluxes once the sun gets above the horizon. The lack of cloud cover over the last couple of summers will undoubtedly have exacerbated this.
- Heating rate is inversely proportional to depth (not volumetric flowrate per se) and is exacerbated by the low summer flowrates of 2025 & 2026.
 - Loss of Ranunculus, in addition to its innumerable impacts on river ecology, will undoubtedly have had a hydrological effect, leading to faster flow speeds and reduced depth for a given volumetric flowrate. How significant this is, I've no idea.
 - Releases of water from the Elan Dam could provide mitigation insofar as they increase average depth, thereby reducing daytime heating rates. But they would need to be large to have any noticeable impact down in Herefordshire.
 - Other, longer-term measures aimed at increasing base flow would have similar beneficial impact on peak water temperatures but, again, it's about scale.